

7 Acknowledgments

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Limitations

By definition, the investigated self-awareness signal is limited to factual recall alone. There are multiple other forms of self-awareness that this work does not address. We look into the rudimentary form of factual recall where all the necessary information (e.g., entity name, entity type, relation) is provided within the immediate query. In open-ended generation tasks, the LM might need to gather this information from scattered context, resolve coreferences, perform multi-hop factual recall implicitly, etc. We leave the investigation of self-awareness under such stressors as a future work. Additionally, the dataset used in this study is restricted in its coverage of entity types and relation categories. Expanding the dataset to include a broader and more diverse range of entities and relational structures would provide a more comprehensive understanding of how self-awareness representations generalize across semantic domains. Finally, while we demonstrate the linearly separable signals of factual self-awareness in the intermediate neural representations and its scaling behavior, it is not known how the model learns to encode this from mere next token prediction training, or the internal causal components that construct and use these signals.

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